KUET · ESE 3123 · Thermo-Fluid Devices

ESE 3123 Question Map

Every question from the 2019, 2020, 2021, 2022, 2023 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 3123 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 3 h for Part A and 3 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×5, 2022 ×4, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). Part A is mostly pumps and flow meters: the venturi/Pitot numericals (A4), affinity laws (A7) and the reciprocating pump numerical (A8) are short and come every year, and each has a calculator. In Part B the heat exchanger numericals (B5, B6) are the safest marks; the normal-shock calculator (B3) gives every downstream property in one go. Leave dimensional analysis (A3) and nozzle derivations (B2) for last.

Section A: Fluid flow, measurement and pumps

Block 1: very likely and easy · 2.1 h

Topic	Chance	Level	Covered
A4 Flow measurement: venturimeter, orifice meter, nozzle meter, Pitot tube, rotameter, turbine meter	Very likely 100%	Easy	22%

A6 Pump performance: characteristic curves, affinity laws, specific speed, cavitation and NPSH, priming, suction height	Very likely 100%	Easy	37%
A8 Reciprocating and other positive-displacement pumps: working, slip, classification, plunger vs diaphragm, numericals	Very likely 100%	Easy	50%
A7 Centrifugal pump numericals: pumps in series, multistage pumps, affinity laws	Very likely 100%	Easy	56%
A5 Centrifugal pump: definition, parts, casings, working, heads and efficiencies, work done by the impeller	Very likely 87%	Easy	67%

Block 2: very likely, medium effort · 0.8 h

Topic	Chance	Level	Covered
A1 Pipe flow theory: laminar and turbulent flow, Reynolds number, entrance region, losses, Darcy-Weisbach, laminar velocity profile	Very likely 100%	Medium	81%

✔ Pass line: 81% of Section A marks covered, enough to choose 3 of the 4 questions (about 2.8 h)

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
A3 Dimensional analysis: dimensions, homogeneity, Rayleigh's method, Buckingham π theorem	Very likely 87%	Hard	88%

Block 4: likely · 1.2 h

Topic	Chance	Level	Covered
A9 Compressors, fans and blowers	Likely 67%	Easy	93%
A2 Pipe flow numericals: laminar head loss and pumping power, pipes in series and parallel, pipe schedules	Likely 60%	Medium	100%

Section B: Compressible flow and heat exchangers

Block 1: very likely and easy · 0.4 h

Topic	Chance	Level	Covered
B5 Heat exchanger LMTD numericals: area, length, outlet temperature, fouling	Very likely 100%	Easy	17%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
B4 Heat exchanger fundamentals: classification, temperature profiles, overall U, fouling factor, LMTD derivations	Very likely 100%	Medium	38%

B1 Compressible flow basics: definitions, Mach number classes, speed of sound, stagnation properties, adiabatic Bernoulli	Very likely 100%	Medium	53%
B3 Shock waves: normal and oblique shocks, Rayleigh line, normal-shock numericals	Very likely 93%	Medium	69%
B6 Effectiveness-NTU method	Very likely 93%	Medium	83%

✔ Pass line: 83% of Section B marks covered, enough to choose 3 of the 4 questions (about 3.4 h)

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B2 Isentropic flow in nozzles and diffusers: area-velocity relation, choking, back pressure, nozzle numericals	Very likely 100%	Hard	100%

Pass plan

Five papers (2019-2023). Normal papers: 210 marks, answer any THREE of four questions in each section (35 marks each). 2020 was a short exam (120 marks, any TWO of three per section). In 2020-2022 Section A = pipe flow, dimensional analysis, flow measurement and pumps; Section B = compressible flow and heat exchangers. **The topics move:** in 2023 Section A holds compressible flow, compressors and pumps while Section B holds measurement, pipe flow and heat exchangers; in 2019 Section A holds pipe flow, measurement and heat exchangers and Section B holds compressible flow, compressors and pumps. So this pack sorts by **topic**, not by question number. **Minimum pass set (by topic):** pipe flow (A1-A2), flow meters (A4), centrifugal pumps (A5-A7), reciprocating pumps (A8), compressible flow and nozzles (B1-B2), heat exchangers (B4-B6). In the paper this means: 2021/2022 A: Q1, Q2, Q3 or Q4; B: Q5, Q7, Q8. 2023 A: Q1, Q3, Q4; B: Q5 or Q6, Q7, Q8. 2019 A: Q1 or Q2, Q3, Q4; B: Q5, Q7, Q8. Answers follow the teachers' material (Fahim sir: pipe flow, dimensional analysis, flow measurement, heat exchanger lectures; Shameem sir: centrifugal and reciprocating pumps, blowers and compressors) and the textbooks they use (Cengel & Cimbala, Bansal, Incropera, Holman). There were no teacher notes for compressible flow, so that unit follows Cengel ch. 12 and Bansal ch. 15. Every numerical was re-checked in Python.

SECTION A: FLUID FLOW, MEASUREMENT AND PUMPS

7 topics

cover 88% of all Section A marks asked

- Pipe flow theory and a laminar-flow numerical (A1, A2)
- Flow meters: venturi/orifice/Pitot derivation and numerical (A4)
- Centrifugal pump: parts, heads, work done, curves, cavitation, a numerical (A5-A7)
- Reciprocating pump: slip, C_d, power (A8); backups: dimensional analysis (A3), compressors and fans (A9)

SECTION B: COMPRESSIBLE FLOW AND HEAT EXCHANGERS

5 topics

cover 85% of all Section B marks asked

- Compressible flow: definitions, speed of sound, stagnation, critical ratios (B1)
- Nozzles: area-velocity relation, choking, a nozzle numerical (B2)
- Heat exchangers: classification, U, fouling, LMTD derivation and numerical (B4, B5)
- Effectiveness-NTU derivation and numerical (B6); backup: normal shocks (B3)

- **Pumps every year (about 2 questions):** centrifugal pump parts and casings, heads and efficiencies, work-done derivation, characteristic curves, cavitation/NPSH, specific speed, and one numerical (pumps in series, multistage, affinity laws); reciprocating pump slip, C_d, power (A5-A8).
- **Heat exchangers every year (about 2 questions):** classification, counter vs parallel, fouling, U, LMTD derivation, effectiveness-NTU derivation, and one LMTD numerical plus one NTU numerical (B4-B6).
- **Compressible flow every year (1-2 questions):** definitions and Mach classes, speed of sound derivation, area-velocity relation, critical ratios, choking/back pressure, a nozzle numerical (B1-B2). Normal shocks (B3) are the next to add.
- **Pipe flow and meters:** laminar vs turbulent, entrance region, major/minor losses, Darcy-Weisbach, laminar profile derivation, a Hagen-Poiseuille numerical; venturi/orifice/Pitot derivation and a numerical (A1, A2, A4). Dimensional analysis (A3) and compressors/fans (A9) are backups.
- **Always draw:** Moody chart sketch, venturimeter, Pitot tube, centrifugal pump with casing, velocity triangles, pump characteristic curves, reciprocating pump indicator diagram, converging-diverging nozzle with pressure plot, temperature profiles of parallel and counter flow.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: FLUID FLOW, MEASUREMENT AND PUMPS: MARKS ASKED PER PAPER

Topic	2019	2020	2021	2022	2023	Years	Marks	
A4 Flow measurement: venturimeter, orifice meter, nozzle meter, Pitot tube, rotameter, turbine meter	57	18	14	39	31	5/5	159	Pass se
A6 Pump performance: characteristic curves, affinity laws, specific speed, cavitation and NPSH, priming, suction height	18	34	17	27	16	5/5	112	Pass se
A8 Reciprocating and other positive-displacement pumps: working, slip, classification, plunger vs diaphragm, numericals	22	18	18	10	27	5/5	95	Pass se
A7 Centrifugal pump numericals: pumps in series, multistage pumps, affinity laws	10	8	12	8	8	5/5	46	Pass se
A5 Centrifugal pump: definition, parts, casings, working, heads and efficiencies, work done by the impeller	15		23	20	19	4/5	77	Pass s
A1 Pipe flow theory: laminar and turbulent flow, Reynolds number, entrance region, losses, Darcy-Weisbach, laminar velocity profile	27	12	20	25	25	5/5	109	Pass set
A3 Dimensional analysis: dimensions, homogeneity, Rayleigh's method, Buckingham π theorem	20		21	6	4	4/5	51	Next to a

A9 Compressors, fans and blowers	18		5	15	3/5	38	Next t
A2 Pipe flow numericals: laminar head loss and pumping power, pipes in series and parallel, pipe schedules	23	15		10	3/5	48	Pass s

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: COMPRESSIBLE FLOW AND HEAT EXCHANGERS: MARKS ASKED PER PAPER

Topic	2019	2020	2021	2022	2023	Years	Marks	
B5 Heat exchanger LMTD numericals: area, length, outlet temperature, fouling	15	22	15	13	32	5/5	97	Pass s
B4 Heat exchanger fundamentals: classification, temperature profiles, overall U, fouling factor, LMTD derivations	20	6	35	33	23	5/5	117	Pass se
B1 Compressible flow basics: definitions, Mach number classes, speed of sound, stagnation properties, adiabatic Bernoulli	23	6	24	11	24	5/5	88	Pass se
B3 Shock waves: normal and oblique shocks, Rayleigh line, normal-shock numericals		12	19	35	20	4/5	86	Next to a
B6 Effectiveness-NTU method		20	20	24	15	4/5	79	Pass se
B2 Isentropic flow in nozzles and diffusers: area-velocity relation, choking, back pressure, nozzle numericals	12	24	27	24	11	5/5	98	Pass s

Section A: Fluid flow, measurement and pumps

topics A1-A9 (usually questions 1-4)

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A4 Flow measurement: venturimeter, orifice meter, nozzle meter, Pitot tube, rotameter, turbine meter

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 159 marks in total

STUDY THESE

- All obstruction meters: Bernoulli + continuity ⇒ $Q_{th} = A_1A_2\sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}$, $Q_{act} = C_dQ_{th}$
- Differential manometer: $h = x(S_h/S_o - 1)$ metres of flowing liquid
- Hydraulic coefficients $C_c = A_c/A_o$, $C_v = V_{act}/V_{th}$, $C_d = C_cC_v$; venturi 0.97-0.99, nozzle 0.95-0.98, orifice 0.61-0.65
- Pitot: $V = C_v\sqrt{(2gh)}$; vena contracta = smallest jet section about D/2 downstream of an orifice; rotameter = variable area float meter; turbine meter = rotor speed ∝ velocity

> 2023 Q5(b) 15 marks
>
> What is an orifice meter? Derive an expression for the flow rate.

> 2023 Q5(c) 6 marks
>
> Relative accuracy of venturi meter, nozzle meter and orifice meter.

> 2023 Q5(d) 10 marks
>
> Same venturimeter as 2022 2(d): flow rate.

> 2022 Q2(b) 6 marks
>
> Commonly used flow-rate measuring instruments; differences between venturimeter and orifice meter.

> 2022 Q2(c) 12 marks
>
> Derive an expression for the volumetric flow rate through a nozzle meter.

> 2022 Q2(d) 11 marks
>
> Horizontal venturimeter, inlet 30 cm, throat 15 cm, differential manometer 20 cm of mercury, $C_d = 0.98$: rate of flow.

> 2022 Q1(c) 10 marks
>
> Submarine “Titan”, same set-up with 190 mm of mercury: speed.

> 2021 Q2(c) 4 marks
>
> Briefly discuss vena contracta.

> 2021 Q2(b) 10 marks
>
> What are hydraulic coefficients? Deduce an expression for the actual velocity at a point using a Pitot tube.

> 2020 Q1(b) 12 marks
>
> Define hydraulic diameter. Deduce an expression for the actual flow rate through a venturimeter.

> 2020 Q1(c) 6 marks
>
> Submarine 15 m below the sea surface; Pitot tube to a mercury U-tube shows 200 mm; $S_{Hg} = 13.6$, $S_{sea} = 1.026$: speed of the submarine.

> 2019 Q6(c) 12 marks
>
> Define venturimeter. Describe it and find an expression for the discharge through a pipe.

> 2019 Q3(b) 12 marks
>
> Derive an expression for the volumetric flow rate through an orifice meter.

> 2019 Q3(a) 12 marks
>
> What is a Pitot tube? How is it used to measure velocity at a point in a pipe or channel?

> 2019 Q3(c) 11 marks
>
> Working principle of a rotameter.

> 2019 Q6(d) 10 marks
>
> Air through a venturimeter, inlet 400 mm, throat 150 mm; inlet 137 kN/m^2 abs, 15 °C; throat 127 kN/m^2 abs; $R = 287$, $\gamma = 1.4$: mass flow rate.

Full answer: ESE 3123 Study Pack, topic A4.

A6 Pump performance: characteristic curves, affinity laws, specific speed, cavitation and NPSH, priming, suction height

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 112 marks in total

STUDY THESE

- Main characteristics (H, P, η vs Q at several speeds), operating characteristics (H, P, η vs Q at the design speed), constant-efficiency (iso-η) curves; operating point = pump curve ∩ system curve

- Affinity laws: $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$ (same pump: $Q \propto N$, $H \propto N^2$, $P \propto N^3$)
- Specific speed $N_s = N\sqrt{Q}/H_m^{3/4}$: speed of a geometrically similar pump giving 1 m^3/s against 1 m; fixes the impeller type
- Cavitation: local $p <$ vapour pressure $\Rightarrow$ bubbles collapse $\Rightarrow$ pitting, noise, fall in H and η; avoid with $NPSH_A > NPSH_R$; $NPSH = p_s/\rho g + V_s^2/2g - p_v/\rho g$
- Priming: fill suction pipe and casing with liquid (head $\propto \rho$, air gives ~1/800 of the head); max suction height $h_s = p_{atm}/\rho g - p_v/\rho g - h_{fs} - V_s^2/2g - NPSH$

> 2023 Q3(a) 8 marks
>
> What are the affinity laws? Illustrate and explain the characteristic curve of a centrifugal pump.

> 2023 Q4(a) 8 marks
>
> Explain cavitation in pump operation and how NPSH is related to cavitation.

> 2022 Q4(b) 8 marks
>
> Significance of the characteristic curves of a centrifugal pump; main and operating characteristic curves.

> 2022 Q4(a) 12 marks
>
> What is specific speed? Show that it depends on the manometric head and the discharge.

> 2022 Q4(c) 7 marks
>
> Define cavitation. Why and where does it occur? Significance of NPSH in the pump industry.

> 2021 Q4(a) 7 marks
>
> What is priming? Why is it necessary?

> 2021 Q4(b) 10 marks
>
> Derive an expression for the maximum suction height of a centrifugal pump.

> 2020 Q3(a) 7 marks
>
> Differences between fans, compressors and blowers. Draw the main and operating characteristic curves of a centrifugal pump.

> 2020 Q3(b) 10 marks
>
> Define specific speed of a centrifugal pump. Deduce the expression for it.

> 2020 Q3(c) 5 marks
>
> What is NPSH? Its significance.

> 2020 Q2(c) 12 marks

What will happen if the impeller of a centrifugal pump is made to rotate in the presence of air?

2019 Q7(b) 10 marks

Discuss pump affinity laws. When does an engineer use them?

2019 Q8(a) 8 marks

Define cavitation. Effects of cavitation and precautions against it.

Full answer: ESE 3123 Study Pack, topic A6.

A8 Reciprocating and other positive-displacement pumps: working, slip, classification, plunger vs diaphragm, numericals

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 95 marks in total

STUDY THESE

- Piston/plunger in a cylinder driven by crank and connecting rod; suction and delivery non-return valves; flow is pulsating
- Q_{th} = ALN/60 (single acting), 2ALN/60 (double acting, rod neglected), × number of cylinders for multi-throw
- Slip = $Q_{th} - Q_{act}$; % slip = $(1 - C_d) \times 100$; negative slip when $Q_{act} > Q_{th}$ (long suction pipe, short delivery, high speed)
- Power = $\rho g Q_{th}(h_s + h_d)$ (theoretical); with losses use $H = H_{st} + h_{fs} + h_{fd} + V_d^2/2g$ and divide by η

2023 Q4(c) 10 marks

Define slip and percentage slip. Using diagrams describe the working of a reciprocating pump.

2023 Q4(b) 8 marks

Advantages and disadvantages of plunger pumps and diaphragm pumps: applications and performance.

2023 Q4(d) 9 marks

Single-acting pump, 50 rpm, stroke 40 cm, piston 25 cm, discharges 0.015 m^3/s: theoretical discharge, slip and % slip, C_d, power.

2022 Q3(a) 10 marks

> Define hydraulic machines. Explain the working principle of a reciprocating pump.

> 2021 Q4(c) 8 marks
> Classification of reciprocating pumps. Differences between centrifugal and reciprocating pumps.

> 2021 Q4(d) 10 marks
> Double-acting pump at 40 rpm discharges 2 m^3/min; stroke 500 mm, piston 250 mm; delivery 18 m, suction 5 m: slip and power.

> 2020 Q2(a) 6 marks
> What is the reason for negative slip in a reciprocating pump?

> 2020 Q2(b) 12 marks
> Single-acting pump, plunger 250 mm, stroke 350 mm, 60 rpm, delivers 16.5 L/s against suction 5 m and delivery 20 m: theoretical discharge, C_d, slip, % slip, power.

> 2019 Q8(b) 12 marks
> Define slip and percentage slip of a reciprocating pump. Describe its working with neat sketches.

> 2019 Q8(c) 10 marks
> Three-throw pump delivers 0.1 m^3/s against 100 m; cylinder 25 cm × 50 cm stroke; friction 1 m suction, 16 m delivery; delivery velocity 1.4 m/s; η 90%, slip 2%: speed and power.

Full answer: ESE 3123 Study Pack, topic A8.

A7 Centrifugal pump numericals: pumps in series, multistage pumps, affinity laws

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 46 marks in total

STUDY THESE

- Specific speed is the same for geometrically similar pumps: $N_1\sqrt{Q_1}/H_1^{3/4} = N_2\sqrt{Q_2}/H_2^{3/4}$ (H per stage)
- Number of stages n = total head/head per stage (round up); diameter from the head coefficient $\sqrt{H}/(DN)$ = constant

- Affinity laws for one pump at a new speed: $N_2 = N_1\sqrt{(H_2/H_1)}$, $Q_2 = Q_1 N_2/N_1$, $P_2 = P_1(N_2/N_1)^3$

> 2023 Q3(d) 8 marks
>
> Pump at 80 gpm, maximum head 90 ftH_2O, 6 hp at 750 rpm; system head rises to 120 ftH_2O: new impeller speed.

> 2022 Q4(d) 8 marks
>
> Double-stage pump, impeller 30 cm, 2000 rpm, lifts 5 m^3/s to 25 m, η 80%: stages and diameter of a similar multistage pump lifting 7 m^3/s to 200 m at 1500 rpm.

> 2021 Q3(d) 12 marks
>
> Single-stage pump, impeller 50 cm, 1800 rpm, lifts 4 m^3/s to 25 m, η 70%: stages and impeller diameter of a similar multistage pump lifting 5 m^3/s to 180 m at 1300 rpm.

> 2020 Q3(d) 8 marks
>
> Water from a deep well, total head 100 m; identical pumps at 900 rpm, specific speed 30, discharge 0.20 m^3/s: number of pumps.

> 2019 Q7(d) 10 marks
>
> Pump at 100 gpm, head 100 ftH_2O, 8 hp at 875 rpm; a new heat exchanger raises the head to 150 ftH_2O: new impeller speed.

Full answer: ESE 3123 Study Pack, topic A7.

A5 Centrifugal pump: definition, parts, casings, working, heads and efficiencies, work done by the impeller

Pass set · Very likely · 87% · Easy

Asked in 4 of 5 papers · 77 marks in total

STUDY THESE

- Pump = hydraulic machine that turns mechanical energy into pressure energy of a liquid; centrifugal pump does it by a rotating impeller (forced vortex) and a casing that turns velocity into pressure
- Parts: impeller (open, semi-open, closed), casing (volute, vortex, with guide blades), suction pipe with foot valve and strainer, delivery pipe with valve, shaft, bearings, stuffing box

- Heads: static $H_s = h_s + h_d$; manometric $H_m = H_s + h_{fs} + h_{fd} + V_d^2/2g$; $\eta_{mano} = gH_m/V_{w2}u_2$, η_{mech} = impeller power/shaft power, $\eta_o = \eta_{mano}\eta_{mech}$
- Work done per unit weight (radial entry $V_{w1} = 0$) = $V_{w2}u_2/g$; $u = \pi DN/60$

> `2023` `Q3(b)` `9 marks`
>
> Define pump. Illustrate and describe the working principle of a single-stage centrifugal pump.

> `2023` `Q3(c)` `10 marks`
>
> Derive an expression for the work done by the impeller on the liquid per second per unit weight.

> `2022` `Q3(c)` `10 marks`
>
> Describe the casing with guide blades of a centrifugal pump with a schematic diagram.

> `2022` `Q3(b)` `10 marks`
>
> Short notes: manometric efficiency, static head, mechanical efficiency, manometric head.

> `2021` `Q3(a)` `5 marks`
>
> What are hydraulic machines? Basic difference between a turbine and a pump.

> `2021` `Q3(b)` `10 marks`
>
> Define centrifugal pump. Main parts, with sketches.

> `2021` `Q3(c)` `8 marks`
>
> Define: mechanical efficiency, manometric head, static head, manometric efficiency.

> `2019` `Q7(c)` `7 marks`
>
> Define pump. Describe with neat sketches the working principle of a centrifugal pump.

> `2019` `Q7(a)` `8 marks`
>
> Define: static suction head, static delivery head, water horsepower, pump efficiency.

Full answer: ESE 3123 Study Pack, topic A5.

A1 Pipe flow theory: laminar and turbulent flow, Reynolds number,

Pass set · Very likely · 100% · Medium

entrance region, losses, Darcy-Weisbach, laminar velocity profile

Asked in 5 of 5 papers · 109 marks in total

STUDY THESE

- Reynolds number Re = ρVD/μ = inertia/viscous force; laminar Re < 2300, transitional 2300-4000, turbulent > 4000; hydraulic diameter $D_h = 4A_c/p$
- Entrance region: boundary layer grows from the wall, irrotational core shrinks; fully developed when the profile stops changing ($L_h \approx 0.05ReD$ laminar, ≈ 10D turbulent)
- Major loss $h_f = f(L/D)(V^2/2g)$ (Darcy-Weisbach); minor losses $h_m = K\,V^2/2g$ (entrance, exit, bends, valves, sudden enlargement/contraction)
- Laminar: $u(r) = 2V_{avg}(1 - r^2/R^2)$, $u_{max} = 2V_{avg}$, $\Delta P = 32\mu LV/D^2$, $f = 64/Re$; Bernoulli from Euler's equation

> 2023 Q6(c) 4 marks
>
> Differentiate between major and minor losses in pipe flow.

> 2023 Q6(a) 6 marks
>
> Short notes: irrotational flow region, hydrodynamic entrance region, hydrodynamically fully developed region.

> 2023 Q6(b) 15 marks
>
> With an appropriate force balance show that for fully developed laminar flow in a circular pipe $u(r) = 2V_{avg}(1 - r^2/R^2)$.

> 2022 Q1(b) 15 marks
>
> What are the different types of losses when a fluid flows through a pipe? Derive the expression for the loss of head due to friction in pipes.

> 2022 Q1(a) 10 marks
>
> Define hydraulic diameter. Show the different flow regions of a fluid flowing through a pipe and how they differ with Reynolds number.

> 2021 Q1(a) 10 marks
>
> What are the different kinds of losses in pipe flow? Briefly discuss.

> 2021 Q1(b) 10 marks
>
> State and discuss Bernoulli's equation. How can it be derived from Euler's equation?

> 2020 Q1(a) 12 marks

> What do you understand by laminar flow and turbulent flow? Derive an expression for the loss of head due to friction in pipes.

> 2019 Q2(a) 12 marks
>
> Briefly discuss: (i) hydrodynamic entrance region and fully developed flow, (ii) minor losses in pipe flow, (iii) working principle of turbine flow meters.

> 2019 Q1(c) 15 marks
>
> With an appropriate force balance show that for fully developed laminar flow in a circular pipe $u(r) = 2V_{avg}(1 - r^2/R^2)$.

Full answer: ESE 3123 Study Pack, topic A1.

A3 Dimensional analysis: dimensions, homogeneity, Rayleigh's method, Buckingham π theorem

Next to add · Very likely · 87% · Hard

Asked in 4 of 5 papers · 51 marks in total

STUDY THESE

- Fundamental dimensions M, L, T (and θ); a dimensionally homogeneous equation has the same dimensions in every term
- Buckingham π: n variables, m fundamental dimensions ⇒ (n − m) independent dimensionless π groups; choose m repeating variables (geometry D, flow V, fluid ρ)
- Rayleigh: assume the variable = K × product of the others with unknown powers and match powers; fine for ≤ 4 variables, Buckingham is systematic for more
- Pipe results: $\tau_w/\rho V^2 = f(Re, \varepsilon/D)$; $\Delta P/\rho V^2 = f(l/D, \nu/VD, k/D)$

> 2023 Q5(a) 4 marks
>
> State Buckingham's π theorem.

> 2022 Q2(a) 6 marks
>
> What is meant by dimensional homogeneity? Discuss with an example.

> 2021 Q2(a) 8 marks
>
> State and discuss Buckingham-π theorem. How does it differ from Rayleigh's method?

> 2021 Q2(d) 13 marks
>
> ΔP in a pipe of diameter D and length l in turbulent flow depends on V, ν, ρ and roughness k. Using Buckingham's π theorem obtain an expression for ΔP.

> 2019 Q1(a) 5 marks
>
> What is meant by qualitative description of a physical quantity? Discuss in brief.

> 2019 Q1(b) 15 marks
>
> Incompressible fluid (ρ, μ) in a long horizontal round pipe of diameter D, average velocity V, roughness ε: develop a non-dimensional relationship between wall shear stress τ_w and the other parameters.

Full answer: ESE 3123 Study Pack, topic A3.

A9 Compressors, fans and blowers

Next to add · Likely · 67% · Easy

Asked in 3 of 5 papers · 38 marks in total

STUDY THESE

- Fan (pressure ratio < 1.11, up to ~1136 mmWC), blower (1.11-1.20), compressor (> 1.20)
- Compressor purposes: compressed air for tools and control, refrigeration, gas transport, combustion air, liquefaction, process reactions
- Positive displacement (reciprocating, rotary) squeezes a trapped volume; dynamic (centrifugal, axial) gives the gas velocity then diffuses it into pressure
- Forced draft fan pushes air into the furnace (cold, clean air, positive furnace pressure); induced draft fan pulls flue gas out (hot, dirty gas, negative pressure)

> 2023 Q2(c) 15 marks
>
> Purpose of a compressor; working principles of centrifugal and reciprocating compressors.

> 2022 Q3(d) 5 marks
>
> Basic differences between fan, compressor and blower.

> 2019 Q6(b) 8 marks
>
> Purpose of a compressor; working principle of centrifugal and reciprocating compressors.

> 2019 Q6(a) 5 marks
>
> Differentiate between a compressor and a blower.

> 2019 Q8(d) 5 marks
>
> Differentiate between forced and induced draft fans.

Full answer: ESE 3123 Study Pack, topic A9.

A2 Pipe flow numericals: laminar head loss and pumping power, pipes in series and parallel, pipe schedules

Pass set · Likely · 60% · Medium

Asked in 3 of 5 papers · 48 marks in total

STUDY THESE

- Check Re first: Re < 2300 ⇒ $f = 64/Re$; $h_L = f(L/D)V^2/2g$; $\Delta P = \rho g h_L$; pumping power $W = \dot{V}\Delta P$
- Series: same Q, losses add; parallel: same head loss, flows add, $Q \propto D^{2.5}$ (same f, same L); $h_f = 8fLQ^2/(\pi^2 g D^5)$
- Pipe schedule: nominal size fixes the outside diameter (3-in pipe OD = 3.500 in); schedule number fixes the wall thickness; flow area from the inside diameter

> 2023 Q6(d) 10 marks
>
> Water at 10 °C (ρ = 999.7 kg/m^3, $\mu = 1.307\times10^{-3}$ kg/m s) flows in a 0.2 cm diameter, 15 m long pipe at 1.2 m/s: pressure drop, head loss, pumping power.

> 2021 Q1(c) 15 marks
>
> Two pipes each 300 m long, diameters 0.30 m and 0.15 m, carry 0.085 m^3/s from a reservoir. Ratio of head lost in series to parallel connection. Neglect minor losses.

> 2019 Q2(b) 14 marks
>
> Water at 5 °C (ρ = 1000 kg/m^3, $\mu = 1.519\times10^{-3}$ kg/m s) flows in a 0.3 cm diameter, 9 m long horizontal pipe at 0.9 m/s: head loss, pressure drop, pumping power.

> 2019 Q2(c) 9 marks
>
> Flow area and wall thickness of: (i) 3-nom sch 10S, (ii) 3-nom sch 40, (iii) 3-nom XXS.

Full answer: ESE 3123 Study Pack, topic A2.

Section B: Compressible flow and heat exchangers

topics B1-B6 (usually questions 5-8)

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B5 Heat exchanger LMTD numericals: area, length, outlet temperature, fouling

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 97 marks in total

STUDY THESE

- Energy balance first: $Q = \dot{m}_c c_{pc}(T_{c,o} - T_{c,i}) = \dot{m}_h c_{ph}(T_{h,i} - T_{h,o})$ gives the missing outlet temperature
- Then ΔT_1, ΔT_2 at the two ends (counter: hot in vs cold out), ΔT_{lm}, $A = Q/(U\Delta T_{lm})$, $L = A/(\pi D)$
- Fouling: $1/U_{dirty} = 1/U_{clean} + R_f$; tube-side h from Dittus-Boelter $Nu = 0.023Re^{0.8}Pr^{0.4}$, laminar annulus Nu from table

> **2023 Q7(c) 20 marks**
>
> Water 20 → 80 °C by oil 160 → 140 °C, D_i 20 mm, U = 500, q = 3000 W: length. After 3 years water leaves at 65 °C: new q, oil outlet, U and water-side fouling factor.

> **2023 Q8(c) 12 marks**
>
> Milk 250 L/h chilled 38.6 → 13 °C by ground water at 10 °C, 0.72 m^3/h; ρ_{milk} 1030, c_p 3860: UA, length for a 50 mm pipe with U = 1000; water outlet temperature.

> **2022 Q7(c) 13 marks**
>
> Water 68 kg/min heated 35 → 75 °C by oil (c_p 1.9 kJ/kg°C) 110 → 75 °C in a counter-flow double pipe; U = 320 $W/m^2°C$: area.

> **2021 Q7(c) 15 marks**
>
> Engine oil 80 → 50 °C, water in at 20 °C; D_i 2.5 cm; $\dot{m}_w$ 0.08, $\dot{m}_{oil}$ 0.16 kg/s; h_w 100, h_{oil} 80 $W/m^2°C$; fouling 0.00018 each; c_w 4180, c_{oil} 2020: tube length.

> **2020 Q5(c) 12 marks**
>
> Counter-flow double tube: water 20 → 80 °C at 1.2 kg/s heated by geothermal water at 160 °C, 2 kg/s; tube 1.5 cm; U = 640 W/m^2K: length.

> **2020 Q6(b) 10 marks**
>
> Counter-flow shell-and-tube: water 0.8 kg/s from 30 to 80 °C; oil 120 → 85 °C; U = 125 $W/m^2°C$: area.

> **2019 Q4(c) 15 marks**

> Gas-turbine lube oil cooler: water 0.2 kg/s in D_i 25 mm, oil 0.1 kg/s in annulus D_o 45 mm; oil 100 → 60 °C, water in at 30 °C: tube length (properties given).

Full answer: ESE 3123 Study Pack, topic B5.

B4 Heat exchanger fundamentals: classification, temperature profiles, overall U, fouling factor, LMTD derivations

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 117 marks in total

STUDY THESE

- Heat exchanger: device transferring heat between two fluids at different temperatures, usually through a wall, without mixing
- Classes: by contact (recuperator, regenerator, direct contact), flow (parallel, counter, cross, multipass), construction (double pipe, shell-and-tube, plate, finned), compactness
- $1/UA = 1/h_iA_i + R_{f,i}/A_i + \ln(D_o/D_i)/2\pi kL + R_{f,o}/A_o + 1/h_oA_o$; thin wall: $1/U = 1/h_i + 1/h_o$
- $Q = UA\Delta T_{lm}$, $\Delta T_{lm} = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$; parallel $\Delta T_1 = T_{h,i} - T_{c,i}$; counter $\Delta T_1 = T_{h,i} - T_{c,o}$

> 2023 Q7(a) 5 marks
>
> What is heat exchange? Its applications.

> 2023 Q8(a) 8 marks
>
> Temperature profiles for counter and parallel flow; why counter flow is more effective.

> 2023 Q7(b) 10 marks
>
> Show that $Q = UA_s\Delta T_{lm}$ for a counter-flow heat exchanger.

> 2022 Q7(a) 8 marks
>
> Classification of heat exchangers; short note on the fouling factor.

> 2022 Q8(c) 11 marks
>
> Derive the overall heat transfer coefficient of a heat exchanger including fouling factors.

> 2022 Q7(b) 14 marks
>
> Derive LMTD for a counter-flow heat exchanger.

> 2021 Q8(a) 6 marks
>
> Define heat exchanger. Classification of heat exchangers.

> 2021 Q8(b) 14 marks
>
> What is effectiveness of a heat exchanger? Show that $U = 1/(1/h_i + 1/h_o)$ for a double-pipe heat exchanger.

> 2021 Q7(a) 15 marks
>
> Show that $Q = UA_s\Delta T_{lm}$ for a parallel-flow heat exchanger.

> 2020 Q5(a) 6 marks
>
> What is the fouling factor? Discuss.

> 2019 Q4(a) 10 marks
>
> Define heat exchanger. Temperature profiles for counter and parallel flow. Why is counter flow more effective?

> 2019 Q4(b) 10 marks
>
> Develop an expression for LMTD of a counter-flow heat exchanger.

Full answer: ESE 3123 Study Pack, topic B4.

B1 Compressible flow basics: definitions, Mach number classes, speed of sound, stagnation properties, adiabatic Bernoulli

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 88 marks in total

STUDY THESE

- Compressible flow: density changes are significant (gases at Ma > 0.3); incompressible: ρ constant (liquids, low-speed gases)
- Ma = V/c; incompressible < 0.3, subsonic < 1, transonic 0.8-1.2, sonic = 1, supersonic 1-5, hypersonic > 5
- $c = \sqrt{(dP/d\rho)_s} = \sqrt{(K/\rho)} = \sqrt{(kP/\rho)} = \sqrt{(kRT)}$
- Stagnation: $T_0 = T + V^2/2c_p$; $T_0/T = 1 + (k-1)Ma^2/2$; $P_0/P = (T_0/T)^{k/(k-1)}$; critical (Ma = 1) $T^*/T_0 = 2/(k+1)$, $P^*/P_0 = 0.5283$ (air)

> 2023 Q1(a) 6 marks
>
> Define: compressible flow, sonic flow, shock waves.

> 2023 Q1(c) 9 marks
>
> Define sound. Prove that c = √(γP/ρ) for one-dimensional isentropic flow of an ideal gas.

> 2023 Q1(b) 9 marks
>
> What is stagnation pressure? Derive an expression for it in compressible flow.

> 2022 Q5(a) 11 marks
>
> Differences between compressible and incompressible flow; classification of compressible flow by Mach number.

> 2021 Q5(a) 6 marks
>
> What is compressible flow? Differentiate between compressible and incompressible flow.

> 2021 Q5(c) 4 marks
>
> Critical properties and critical ratios; their significance.

> 2021 Q5(d) 14 marks
>
> Air in a duct: stagnation 100 kPa abs and 35 °C, static 52 kPa abs. Velocity considering and ignoring compressibility. k = 1.4, R = 287.

> 2020 Q4(b) 6 marks
>
> Differentiate between compressible and incompressible flow.

> 2019 Q5(a) 4 marks
>
> Define: subsonic flow, Mach number.

> 2019 Q5(c) 10 marks
>
> Prove that the velocity of sound in a compressible fluid is c = √(K/ρ), K = bulk modulus.

> 2019 Q5(b) 9 marks
>
> Derive Bernoulli's equation for compressible flow when the process is adiabatic.

Full answer: ESE 3123 Study Pack, topic B1.

B3 Shock waves: normal and oblique shocks, Rayleigh line, normal-shock numericals

Next to add · Very likely · 93% · Medium

Asked in 4 of 5 papers · 86 marks in total

STUDY THESE

- Shock wave: very thin (~0.1 µm) region where a supersonic flow suddenly becomes subsonic (normal) or turns (oblique); irreversible, entropy rises, P_0 falls, T_0 constant
- Normal shock relations: $Ma_2^2 = (Ma_1^2 + 2/(k-1))/(2kMa_1^2/(k-1) - 1)$; $P_2/P_1 = (1 + kMa_1^2)/(1 + kMa_2^2)$; $T_2/T_1 = (2 + (k-1)Ma_1^2)/(2 + (k-1)Ma_2^2)$
- Rayleigh line: states with the same mass flux and momentum (continuity + momentum); Fanno line: same mass flux and energy; a normal shock jumps between their intersections

> `2023` `Q2(a)` `8 marks`
>
> Show that the Rayleigh line is $P_2/P_1 = (1 + kMa_1^2)/(1 + kMa_2^2)$.

> `2023` `Q2(b)` `12 marks`
>
> Same as 2022 6(c) but helium (k = 1.667, R = 2.0769 kJ/kg K).

> `2022` `Q6(a)` `6 marks`
>
> Short notes: shock wave, normal shock wave, oblique shock wave.

> `2022` `Q6(b)` `17 marks`
>
> Show that $Ma_2^2 = (Ma_1^2 + 2/(k-1))/(2kMa_1^2/(k-1) - 1)$ across a normal shock.

> `2022` `Q6(c)` `12 marks`
>
> Air, normal shock at Ma = 2.5, upstream 61.64 kPa and 262.15 K: downstream pressure, temperature, velocity, Mach number, stagnation pressure.

> `2021` `Q6(b)` `5 marks`
>
> Differentiate between normal and oblique shock waves.

> `2021` `Q6(c)` `14 marks`
>
> Air enters a C-D nozzle at 1 MPa, 300 K, low velocity; normal shock at the exit plane at Ma = 2: pressure, temperature, Mach number, velocity after the shock.

> `2020` `Q5(b)` `12 marks`
>
> Normal shock in air: upstream Ma 2.2, 106 bar, 16 °C: downstream Mach number, pressure, temperature (k = 1.4).

Full answer: ESE 3123 Study Pack, topic B3.

B6 Effectiveness-NTU method

Pass set · Very likely · 93% · Medium

Asked in 4 of 5 papers · 79 marks in total

STUDY THESE

- Effectiveness $\varepsilon = Q/Q_{max}$, $Q_{max} = C_{min}(T_{h,i} - T_{c,i})$; $NTU = UA/C_{min}$; $C_r = C_{min}/C_{max}$
- Parallel: $\varepsilon = [1 - e^{-NTU(1+C_r)}]/(1 + C_r)$; counter: $\varepsilon = [1 - e^{-NTU(1-C_r)}]/[1 - C_r e^{-NTU(1-C_r)}]$
- Use NTU when outlet temperatures are unknown (rating problem); counter flow always gives the higher ε for the same NTU and C_r

> 2023 Q8(b) 15 marks
>
> Same derivation as 2020 6(a).

> 2022 Q8(a) 6 marks
>
> What is effectiveness of a heat exchanger? Its significance.

> 2022 Q8(b) 6 marks
>
> Justify: a counter-flow heat exchanger has higher effectiveness than a parallel-flow one under the same conditions.

> 2022 Q8(d) 12 marks
>
> Counter flow, A = 12.5 m^2: oil (2000 J/kg°C) at 100 °C, 2 kg/s; water (4170) at 20 °C, 0.48 kg/s; U = 400: water outlet temperature and Q.

> 2021 Q7(b) 5 marks
>
> Physical significance of the NTU method.

> 2021 Q8(c) 15 marks
>
> Counter flow: oil 0.55 kg/s (c_p 2.45 kJ/kg°C) 115 → 40 °C, water 15 → 75 °C, U = 1450: by NTU, water flow, effectiveness, area.

> 2020 Q6(a) 20 marks
>
> Show that $\varepsilon = [1 - e^{-NTU(1 + C_{min}/C_{max})}]/(1 + C_{min}/C_{max})$ for parallel flow.

Full answer: ESE 3123 Study Pack, topic B6.

B2 Isentropic flow in nozzles and diffusers: area-velocity relation, choking, back pressure, nozzle numericals

Pass set · Very likely · 100% · Hard

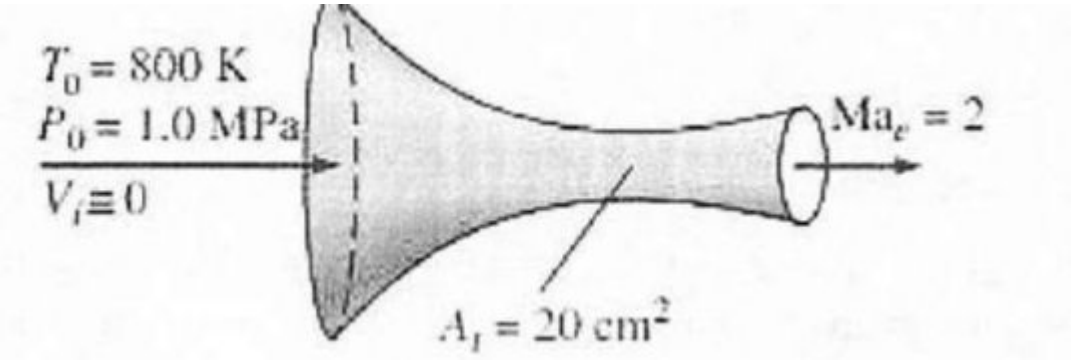
Asked in 5 of 5 papers · 98 marks in total

STUDY THESE

- $dA/A = (Ma^2 - 1)dV/V$: subsonic ⇒ area and velocity change in opposite directions; supersonic ⇒ same direction; Ma = 1 only at the throat
- Converging-diverging duct: nozzle if the flow speeds up through Ma = 1 at the throat; diffuser if supersonic flow is slowed
- Choked flow: $P_b \le P^*$ ⇒ Ma = 1 at the throat and $\dot{m}_{max} = A^*P_0\sqrt{(k/RT_0)}(2/(k+1))^{(k+1)/2(k-1)}$
- $A/A^* = (1/Ma)[(2/(k+1))(1 + (k-1)Ma^2/2)]^{(k+1)/2(k-1)}$; use isentropic tables or formulas

2023 Q1(d) 11 marks

Air enters a C-D nozzle at 1.0 MPa, 800 K, negligible velocity, k = 1.4; exit Ma = 2, throat 20 cm^2: throat conditions, exit conditions and exit area, mass flow rate.



2022 Q5(b) 12 marks

Effect of back pressure on mass flow rate and pressure distribution along a converging nozzle, with sketch.

2022 Q5(c) 12 marks

Air enters a nozzle at 0.2 MPa, 350 K, 150 m/s, isentropic: pressure and temperature where V = speed of sound; area ratio to the entrance.

2021 Q5(b) 11 marks

Show mathematically that a converging-diverging duct can be used as a nozzle or a diffuser.

2021 Q6(a) 16 marks

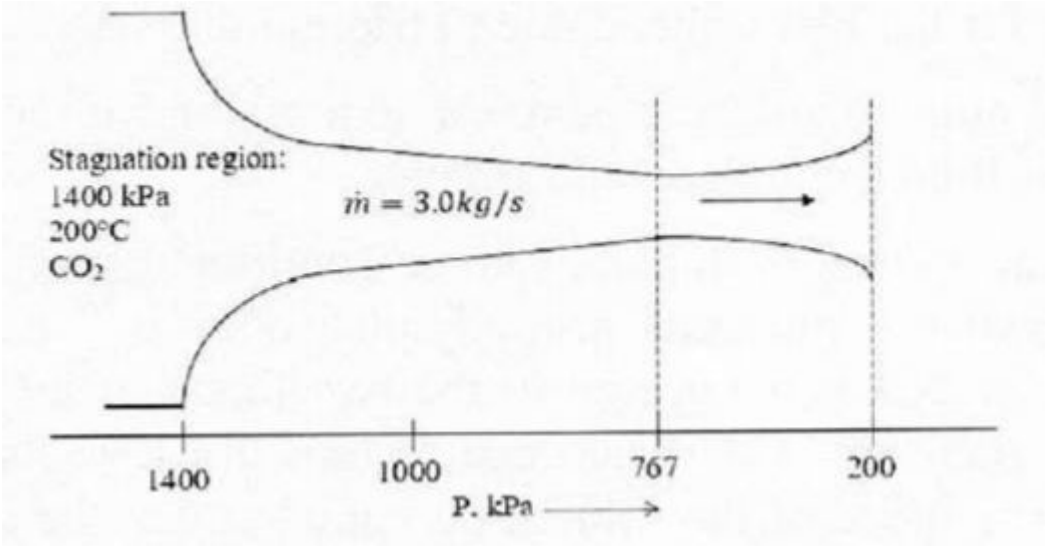
Prove $\dot{m}_{max} = A^*P_0\sqrt{(k/RT_0)}(2/(k+1))^{(k+1)/[2(k-1)]}$.

2020 Q4(a) 12 marks

Prove that velocity and flow area change in the same direction for supersonic isentropic flow.

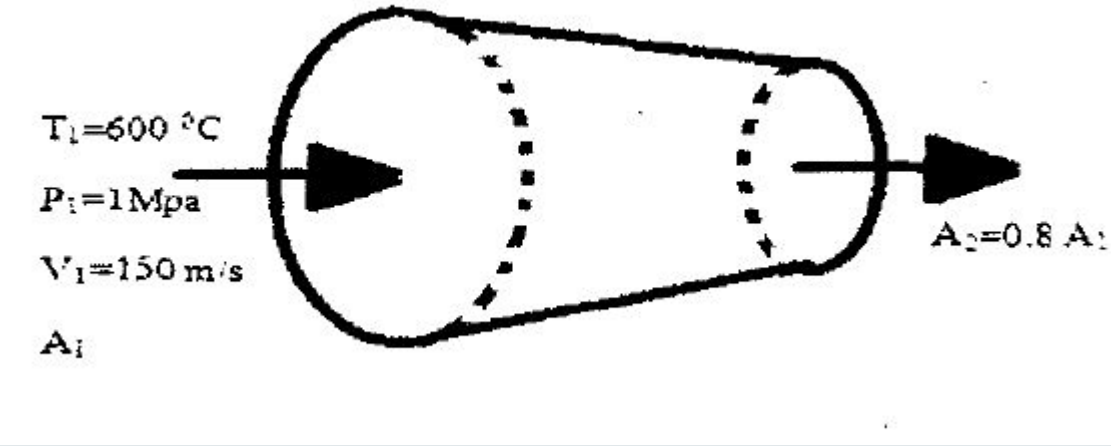
2020 Q4(c) 12 marks

CO_2, 3.00 kg/s, enters at 1400 kPa, 200 °C, low velocity, expands to 200 kPa isentropically: density, velocity, area and Mach number where P = 1200 kPa. c_p = 0.846, k = 1.289, R = 0.1889 kJ/kg K.



2019 Q5(d) 12 marks

Air at 1 MPa, 600 °C (150 m/s in the figure) enters a converging nozzle, throat 50 cm^2: mass flow rate for back pressure (i) 0.7 MPa, (ii) 0.4 MPa.



Full answer: ESE 3123 Study Pack, topic B2.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
AODD	air-operated double diaphragm (pump)
ASME	American Society of Mechanical Engineers

BEP	best efficiency point
BHP	brake horsepower
C-D	converging-diverging
FD	forced draught
gpm	gallons per minute
hp	horsepower
HX	heat exchanger
ID	induced draught (or inside diameter)
kPa	kilopascal
LMTD	log mean temperature difference
MPa	megapascal
NPSH	net positive suction head
NTU	number of transfer units
rpm	revolutions per minute
TEMA	Tubular Exchanger Manufacturers Association
WHP	water horsepower
XXS	double extra strong (pipe schedule)

Built from the scanned ESE 3123 papers (2019, 2020, 2021, 2022, 2023). Question text is cleaned up from the scans.